Nachiketa Gulati

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Portfolio: https://nachi-g.github.io/ | **GitHub**: https://github.com/Nachi-G

EDUCATION

The University of Texas at Austin, Expected May 2030
B.S. in Mechanical Engineering | Intended Robotics Minor
Relevant Fall Coursework: *Statics, Differential Equations with Linear Algebra, Intro to Eng. Design & Graphics*

The Woodlands High School, 2026
Rank 4/1072 | GPA: 4.0/4.0 | SAT: 1560
Coursework: Dual Enrollment Linear Algebra & Calc 1-3, AP Physics C (Mech & E&M), AP Statistics, AP CSA

EXPERIENCE

LearnToBot, Robotics Engineer, April 2026 - Present **https://nachi-g.github.io/projects**
- Featured projects: voice-controlled ESP32-S3 rover integrating Whisper STT + Gemma LLM pipeline; PID-controlled self-balancing robot; OpenCV Soccer Goalie with real-time trajectory tracking
- Developed 10+ robotics projects (CAD, electronics, code) for 6+ camps across 3 cities; created comprehensive assembly manuals, wiring diagrams, and instructional documentation.
- Rapidly prototyped in SolidWorks under design constraints, covering component selection, circuit design, power distribution, DFM for 3D printing, and BOM cost analysis.

Head of Growth & Head Coach, KEB Tennis Camps, April - July 2026 **Camp Instagram**
- **Tripled revenue**, expanding camp to **190+** campers
- Generated **25K+ impressions** leading growth campaigns (flyers, referral program, social media content)
- Scheduled and **managed 10 assistant coaches**; developed level-based curriculum to guide daily instruction

PROJECTS

ShinSavers: An Experimental Approach to Alleviating Medial Tibial Stress Syndrome, 11/2024 - 4/2025
- Designed and prototyped a biomechanical device intended to attenuate impact forces during running
- Developed a treadmill-based experiment using accelerometers to repeatably collect impact data
- Wrote Python scripts to preprocess experimental data for comparative analysis across device materials

RetiBox, Technical Lead and Founder, 5/2025 - 2/2026 **Demo Video**
- Founded and led a 3-person team to build a browser-based Augmented Reality (AR) platform enabling users to create AR models from pictures and share those without downloads
- Built functional prototype utilizing HTML, CSS, Python, Django, and deployed on AWS Amplify

SKILLS

Programming: Python, CircuitPython, ESP32, Raspberry Pi, OpenCV, Git, HTML/CSS, Matplotlib, C++ (basic)
CAD & Prototyping: SolidWorks, Onshape, 3D Printing, Fritzing/Cirkit, Rapid Prototyping
Certifications: CSWA, Supervised Machine Learning (Pursuing Andrew Ng's ML Specialization)

ACCOMPLISHMENTS

- UIL 6A **State Champion** Team Tennis (2025)
- Selected for Math Bowl Calculus Team **(2 of ~200 students)**
- Rice University Young Scholars Invent - 3rd Place (3rd/25 teams)